PROJECT SYNA

Offline P2P Messenger & Friend Finder

Off-Grid Location Sharing, Dynamic Radar View & Encrypted P2P Messaging

MOTIVATION & PAIN POINTS

Communication Fails When You Need It Most

Zero Internet Coverage

Hiking, festivals, or natural disasters sever standard cellular connectivity.

Abstract Coordinate Sharing

Raw GPS numbers are useless without active cellular maps and cloud services.

Direction Blindness

Standard map dots fail to visually guide orientation in unfamiliar outdoor terrain.

THE SOLUTION

Syna: Off-Grid Directional Connectivity

Offline P2P Chat

End-to-end encrypted direct messaging between nearby Android devices.

Local Map Logs

Offline tile rendering with last-known coordinates persisted locally in Room DB.

Radar Compass View

360° directional radar with animated ambient edge-glow guidance.

VALUE PROPOSITION

How Syna Differs From Mesh Messengers

Feature Dimension	Traditional Mesh Apps	Syna P2P Radar & Chat
Primary Focus	Text broadcasting & crisis messages	Directional tracking + off-grid chat
UI Experience	Standard 2D chat list view	Dynamic 360° Compass Radar View
Sensor Scope	Network radio radio only	GPS Hardware + Magnetometer
Data CRUD Scope	Store & forward messages	Local waypoint logs + live coordinates

SYSTEM ARCHITECTURE

Android Tech Stack & System Resources

Core Framework

Kotlin, Jetpack Compose, Material 3, MVVM Architecture.

Off-Grid Transport

Google Nearby Connections API (Wi-Fi Direct / BLE).

Cryptography & Auth

Android Keystore, ECDH key exchange, ZXing QR scanner.

Data Persistence

Room SQLite (Chat history, profiles, local waypoints).

Offline Mapping

MapLibre SDK pre-packaged with offline tile packages.

Sensors & Radar

Rotation Vector Sensor + Compose Canvas Ambient Edge Glow.

TEAM COMPOSITION

Role Distribution & Responsibilities

MEMBER 1 (Rafael)

Networking & Security

Nearby Connections API integration, P2P discovery, device handshake, public-key encryption, and background GPS sensor streams.

MEMBER 2 (Andrew)

Database & Logic

Room schema architecture, profile & contact CRUD, message persistence, QR key management, distance and bearing mathematical logic.

MEMBER 3 (Ahsan)

Frontend & UI/UX

Jetpack Compose screens, QR scanner UI, chat interface, offline MapLibre integration, and animated compass radar UI.

PROJECT ROADMAP

4-Week Implementation Plan

WEEK 1 (Nov 4-10)

Foundation

Project architecture, Room schema, wireframes, and raw offline P2P payload exchange test.

WEEK 2 (Nov 11-17)

Messaging

QR identity exchange, authenticated encrypted chat, local Room persistence, and contact CRUD.

WEEK 3 (Nov 18-24)

Location

GPS sharing stream, MapLibre tile loading, distance/bearing math, and Compass Radar view.

WEEK 4 (Nov 25-Dec 1)

Integration

End-to-end testing, error handling, permission edge cases, and live demo stabilization.



Thank You

Questions & Feedback on SYNA Project Proposal